Configuration and secrets

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

THE CONNECTED WEEK

One complete configuration and diagnostic record.

Keep input roles, required fields, public output fields, observed checks and remaining limits together.

Keep one habit: Recall → W1–W3/W5 → independent output-channel case → project startup contract → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Typed contracts and schemas. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Requirements and service boundaries. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- **Configuration** supplies non-**secret** settings; **secret**s provide privileged values. Keep **secret**s out of source, logs and client responses.

- The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**. Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**.

Typed contracts and schemas

> **My remembered idea, example and correction**

Requirements and service boundaries

> **My remembered idea, example and correction**

Lesson reference for workbook practice

W1 · Explain the current example

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main scenario: missing DATA_VERSION triggers a debug dump of all environment values. Repair: validate declared required inputs at startup and report missing key names without values; do not substitute a public default secret. Mock contract: nonblank DATA_VERSION, exact positive integer MAX_WORDS, separate nonblank SIGNING_SECRET placeholder. Public ready record includes only status, data_version, maximum_words.

  Blocked record names missing/invalid keys without their values. Internal state may contain placeholder but must not be serialized into emitted records. Lab reads only supplied dicts, never real environment values, stores or accounts. Tests check both mock private values absent from chosen serialized records. Separate picture public maximum_words80 changes to60 while credential stays unchanged.

  This is a configured value, with no generated answer to prove length enforcement. Protected storage alone does not prove no leaks elsewhere, credential strength, rotation or deployed correctness.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main scenario: missing DATA_VERSION triggers a debug dump of all environment values. Repair: validate declared required inputs at startup and report missing key names without values; do not substitute a public default secret. Mock contract: nonblank DATA_VERSION, exact positive integer MAX_WORDS, separate nonblank SIGNING_SECRET placeholder. Public ready record includes only status, data_version, maximum_words.

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  This is a configured value, with no generated answer to prove length enforcement. Protected storage alone does not prove no leaks elsewhere, credential strength, rotation or deployed correctness.

- A startup script prints every environment variable while debugging a missing data version. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

W3 · Repair a misconception

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main scenario: missing DATA_VERSION triggers a debug dump of all environment values. Repair: validate declared required inputs at startup and report missing key names without values; do not substitute a public default secret. Mock contract: nonblank DATA_VERSION, exact positive integer MAX_WORDS, separate nonblank SIGNING_SECRET placeholder. Public ready record includes only status, data_version, maximum_words.

  Blocked record names missing/invalid keys without their values. Internal state may contain placeholder but must not be serialized into emitted records. Lab reads only supplied dicts, never real environment values, stores or accounts. Tests check both mock private values absent from chosen serialized records. Separate picture public maximum_words80 changes to60 while credential stays unchanged.

  This is a configured value, with no generated answer to prove length enforcement. Protected storage alone does not prove no leaks elsewhere, credential strength, rotation or deployed correctness.

- A report omits this qualification: A classroom mock is not a production credential-management design; no real **secret**s are needed here.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main scenario: missing DATA_VERSION triggers a debug dump of all environment values. Repair: validate declared required inputs at startup and report missing key names without values; do not substitute a public default secret. Mock contract: nonblank DATA_VERSION, exact positive integer MAX_WORDS, separate nonblank SIGNING_SECRET placeholder. Public ready record includes only status, data_version, maximum_words.

  Blocked record names missing/invalid keys without their values. Internal state may contain placeholder but must not be serialized into emitted records. Lab reads only supplied dicts, never real environment values, stores or accounts. Tests check both mock private values absent from chosen serialized records. Separate picture public maximum_words80 changes to60 while credential stays unchanged.

  This is a configured value, with no generated answer to prove length enforcement. Protected storage alone does not prove no leaks elsewhere, credential strength, rotation or deployed correctness.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

Try alone · Selected transfer

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Server **secret** exists, but a browser error includes its full value. What failed and what record should be produced instead?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply a new case independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the service evidence

> Preserve the contract, inputs, observed outcomes and limits.

Complete task and evidence

- Add a labelled **configuration** and **secret**s **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of **configuration** and **secret**s.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Trace the worked case for **Configuration** and **secret**s. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for **Configuration** and **secret**s. Proposed explanation: "The worked result establishes every future case". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Server **secret** exists, but a browser error includes its full value. What failed and what record should be produced instead?

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token.

- maximum_words=80 is public **configuration**. A separate protected store holds a placeholder credential.

- Point to the public setting and the protected credential. What changes when maximum_words becomes 60, and which information must never appear in the startup log?

- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.



Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ **Reference · Project milestone**

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Grounded Learning Service
- Current milestone: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.
- For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token.
- Which labelled step matches this detail? Record startup success without printing the token.
- 1. Load credential
- 2. Log safely
- 3. Load setting

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> | |
> |---|
> | |
> | |
> | |
> | |

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Log safely Record startup success without printing the token. **Configuration** controls behaviour. A credential grants access and needs different storage and handling.

Changed-case question and optional hint

- Changed-case practice: Change maximum_words to 60. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. The configured answer limit becomes 60; the private token stays unchanged.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: The configured answer limit becomes 60; the private token stays unchanged. An ordinary length setting does not require replacing the credential.

Repair and retain the record

Review Typed contracts and schemas (ai-6/3). Goal: Explain the **mechanism** and terminology of typed contracts and schemas. Short practice: Compare A's count 4 with B's count "4".

Which record passes, which gate rejects B, and why does changing A's count to -1 create a different failure?. Return to this week and explain what you changed.

Review Requirements and service boundaries (ai-6/1). Goal: Explain the **mechanism** and terminology of requirements and service boundaries. Short practice: Split the museum visitor's request into two operations.

Which can the service perform for P, and what happens to each operation when Q has no approved card?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Configuration and secrets: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

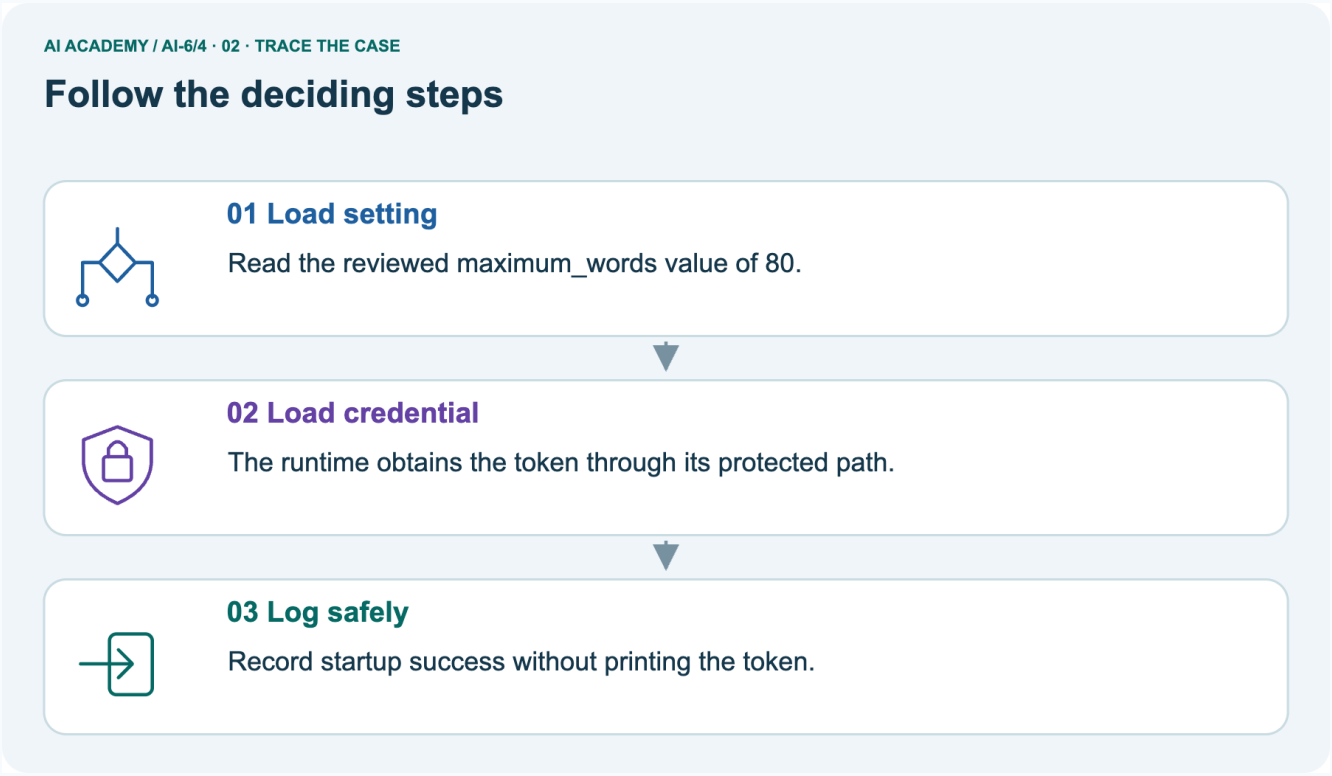
Settings and a sealed credential. A classroom bot needs a public answer-length setting and a private access token. Supplied facts: maximum_words=80 is public **configuration**.

A separate protected store holds a placeholder credential.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



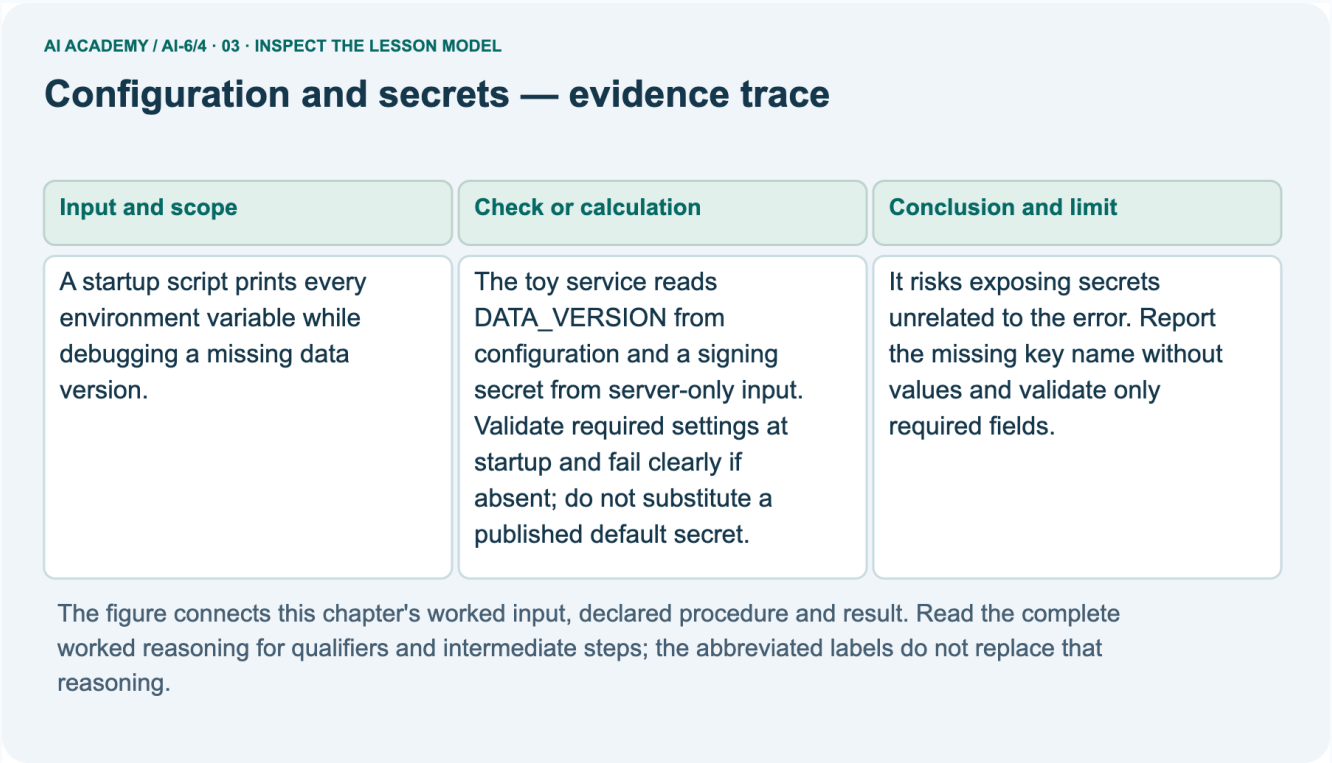
Follow the labelled picture with the matching explanation. Open the vector diagram

Load setting: Read the reviewed maximum_words value of 80. → Load credential: The runtime obtains the token through its protected path. → Log safely: Record startup success without printing the token.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Configuration and secrets — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Configuration and **secret**s — **evidence** trace.

A startup script prints every environment variable while debugging a missing data version. | The toy service reads DATA_VERSION from **configuration** and a signing **secret** from server-only **input**.

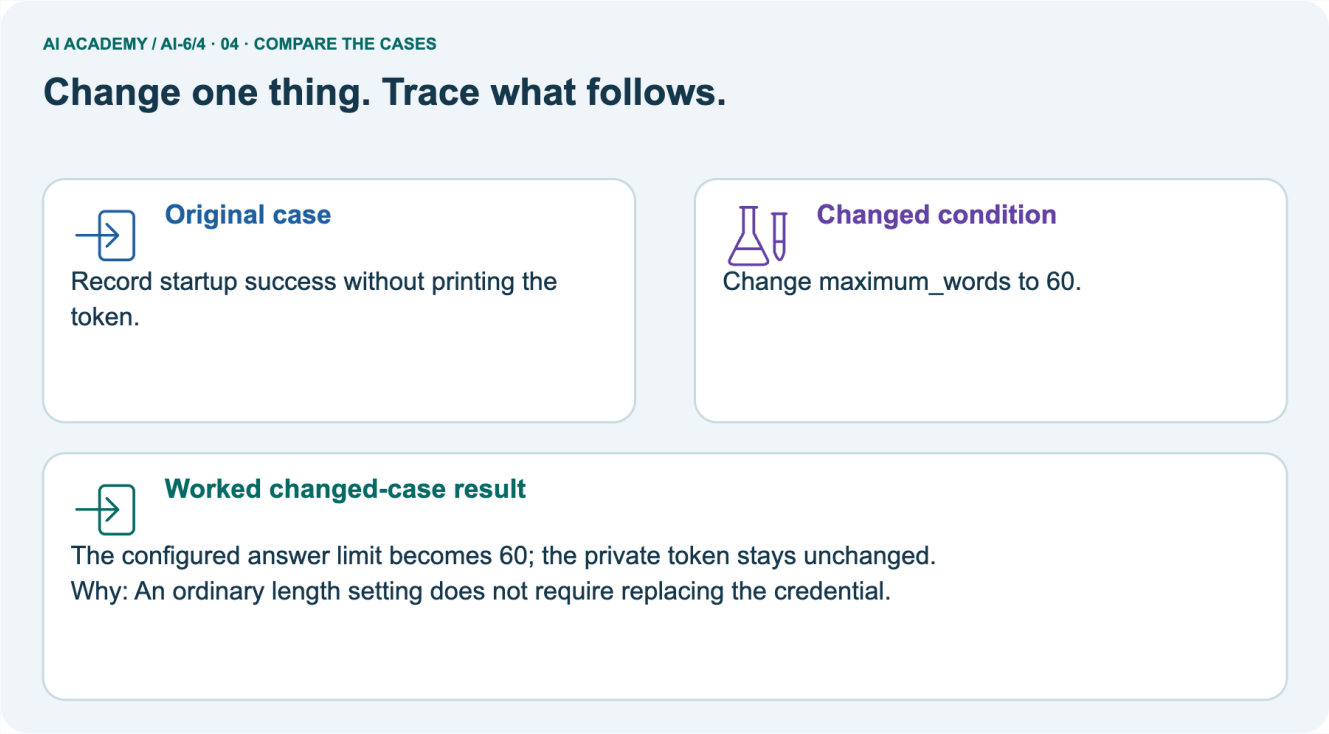
Validate required settings at startup and fail clearly if absent; do not substitute a published default **secret**. | It risks exposing **secret**s unrelated to the error.

Report the missing key name without values and validate only required fields.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

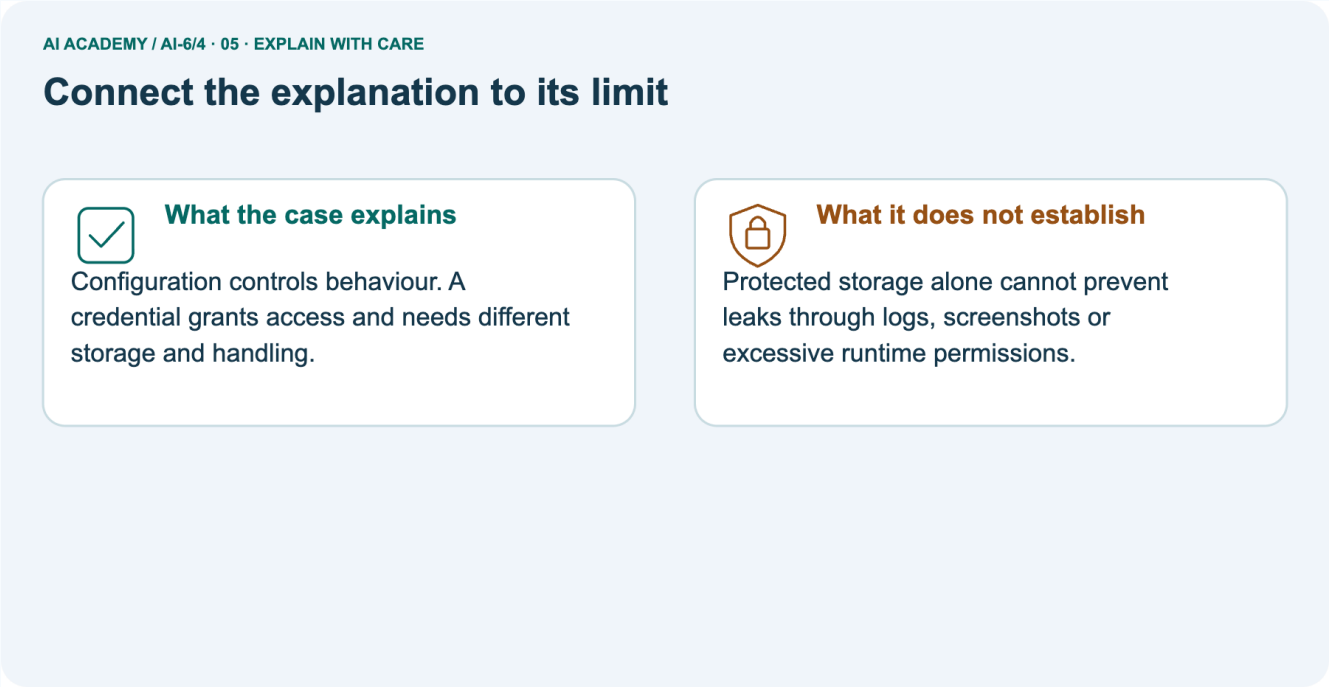
Original case: Record startup success without printing the token. Changed condition: Change maximum_words to 60. Expected result: The configured answer limit becomes 60; the private token stays unchanged.

Reason: An ordinary length setting does not require replacing the credential.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



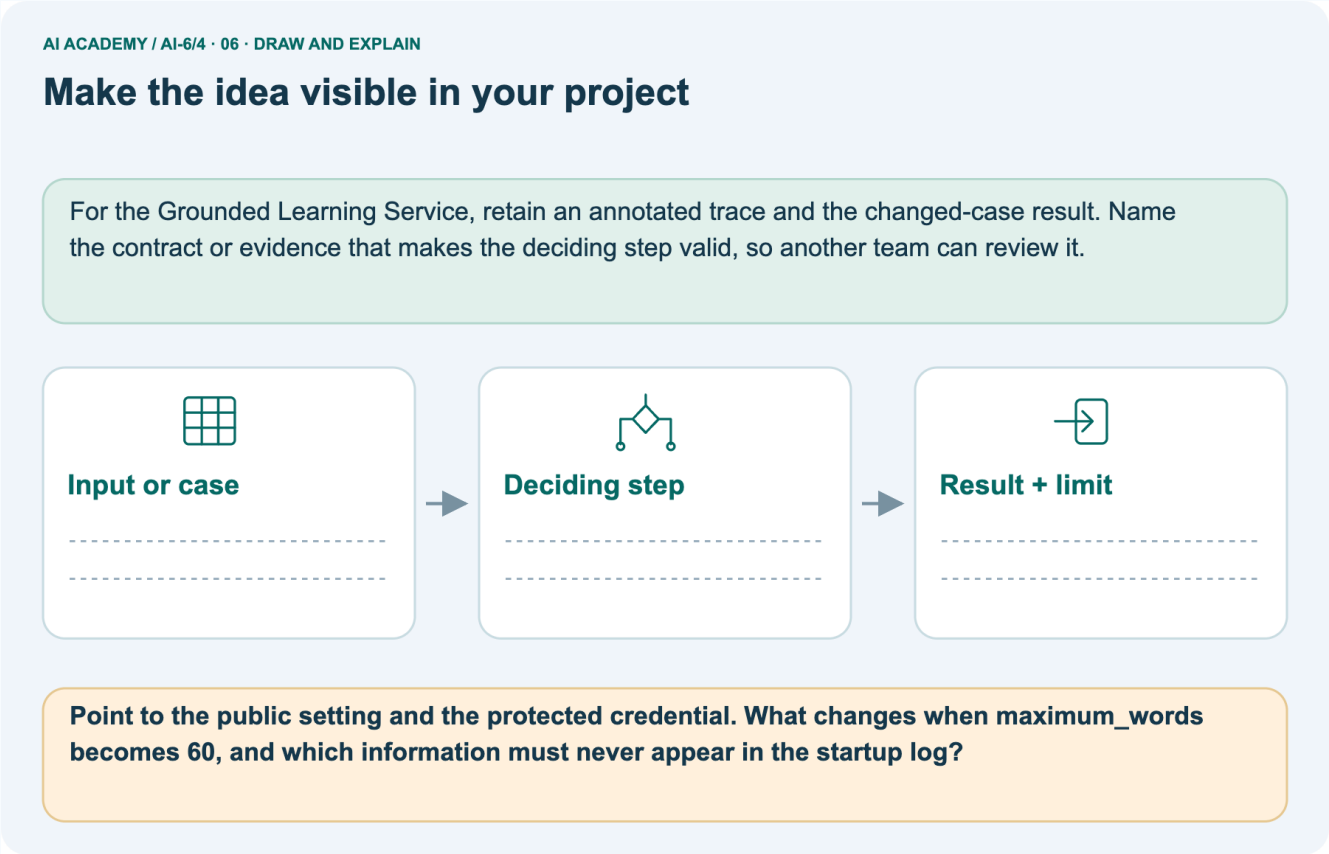
Follow the labelled picture with the matching explanation. Open the vector diagram

Configuration controls behaviour. A credential grants access and needs different storage and handling. Limit: Protected storage alone cannot prevent leaks through logs, screenshots or excessive runtime permissions.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Point to the public setting and the protected credential.

What changes when maximum_words becomes 60, and which information must never appear in the startup log?

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Configuration and secrets: the case

02. Follow the deciding steps

03. Configuration and secrets — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.